Course Code : ECT 456-2 **KQLR/RS – 24 / 1315**

Eighth Semester B. Tech. (Electronics and Communication Engineering) Examination

COMPUTER VISION

Time : 3 Hours] [Max. Marks : 60

Instructions to Candidates :—

(1) All questions are compulsory.
(2) Assume suitable data wherever necessary.
(3) Wherever necessary, illustrate your answer with neat sketches.

1. (a) Elaborate the various applications of Computer Vision with justified example. 5(CO1)

(b) What are the advantages of using Lens over Pinhole camera and which factors are responsible for aperture of lens ? 5(CO1,2)

2. (a) Elaborate the concept of Radiometry and derive the equations for 2D and 3D solid angle with labeled diagram. 5(CO2)

(b) Define the following terms :

(a) Surface Radiance.

(b) Depth of Field. 5(CO2)

3. (a) Explain the types of Image Sensors with neat diagram. 6(CO3)

(b) Derive the equations for Image formation in Stereo Vision in X, Y, Z coordinates with the help of diagram. 4(CO3)

4. (a) Elaborate the process of Edge Detection in Computer Vision with neat diagram. 5(CO4)

(b) Derive the equations for Gradient Edge Detector with diagram and elaborate important features of Gradient Edge Detector. 5(CO4)

5. (a) Explain the concept of Linear Regression Process using Machine Learning with algorithm. 6(CO5)

(b) How does decision function work on Image formation using Machine learning concepts, explain in brief. 4(CO5)

6. Elaborate the concept of pattern classification using Artificial Neural Network with the help of algorithm. 10(CO5)